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Identifying optimal combinations of synchronous condensers for minimum grid strength compliance

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ABSTRACT

This paper proposes a method for the automatic optimization of the size and location of synchronous condensers (SCs) to meet a minimum grid strength requirement that enables the high penetration of inverter-based resources (IBRs) in a weak network. The grid strength is quantified by the effective short-circuit ratio (ESCR), which is an heuristic index that accounts for the voltage interactions between IBRs. The basin hopping algorithm (BHA) is used in combination with the Nelder–Mead algorithm to minimize the total additional short-circuit contribution from the SCs, subject to a minimum ESCR constraint. It is shown that this optimization problem has several solutions with the minimum values of the objective function lying within a narrow range but different allocation of the SCs' capacities to the candidate buses. Therefore, the BHA is iteratively applied to identify the area of the search space in which the optimum solutions are located, and additional criteria are proposed to rank the solutions based on cost and reliability attributes. The effectiveness of the proposed approach is shown using a modified 39-bus New England benchmark system in which conventional synchronous generators were replaced by IBRs, thus leading to weak network conditions.

1. Introduction

The accelerated decarbonization pathways to reach net zero targets in power and energy systems introduces a wide array of technologies interfaced with the grid through power electronics [1]. These inverter-based resources (IBRs), such as wind and solar photovoltaic (PV) generation and battery energy storage systems, are displacing the synchronous equipment that traditionally guaranteed the grid operability [2]. The vast majority of IBRs currently in service, including line-commutated HVDC converters, utilize grid-following (GFL) controls conceived to operate under strong grid conditions [3]. In this scenario, the massive deployment of GFL IBRs can only be accomplished with the aid of appropriate enabling technologies.

The term "grid strength" addresses a broad range of power system operational challenges, including stability, protection and quality of supply [4]. In the context of this paper, the grid strength at a given IBR bus qualitatively refers to the sensitivity of the bus voltage to a variation in the IBR current injection. In a strong point of connection (PoC), the voltage is relatively insensitive to changes in the IBR current injection, whilst this sensitivity is higher in a weak PoC [5].

Several indicators have been defined to quantify the system strength at the PoC of an IBR, the most widely used being the short-circuit ratio (SCR). However, the SCR neglects the potential interactions between IBRs connected electrically close to each other. Therefore, other indicators such as the weighted short-circuit ratio (WSCR), composite short-circuit ratio (CSCR) and the equivalent short-circuit ratio (ESCR) were defined [6]. Out of these, only the ESCR accounts for weak electrical couplings between IBRs [5].

A power-flow model is used in [7] as the mathematical framework to derive the nodal voltage interaction factor, which is an analytic equivalent of the empirical ESCR for small disturbances that enables a rigorous treatment of voltage interactions in multi-infeed HVDC systems. The potential current-saturated operation of power converters is not explicitly considered in the original ESCR formulation but is especially critical in short-circuit analyses as several converters might be operated in this condition during faults in the network [8]. Nonetheless, the ESCR remains an efficient index for the systematic screening of the grid strength in online applications and planning studies to identify conditions with potential power system operability issues, as already reported in Texas [9] and Australia [10].

The increasing penetration of IBRs tends to decrease the ESCR because IBRs have a relatively low short-circuit power contribution (up to or slightly above their nominal current) compared to the synchronous generation they typically replace (around five times their nominal current in the subtransient period). Mitigating measures to avoid curtailing maximum power extraction from solar PV and wind power plants in

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weak networks include the installation of synchronous condensers (SCs) and IBRs with grid-forming controllers (such as virtual synchronous machines) [3]. While SCs are a mature technology that has been used for several decades for the power system stabilization [11], IBRs with grid forming controllers have recently started the path towards commercial deployment in large-scale applications [12].

Metaheuristic algorithms offer efficient solutions for the transmission expansion planning (TEP) [13] and optimal equipment placement, including Flexible AC Transmission System (FACTS) devices [14], distributed generation and capacitor banks [15]. The TEP aims at minimizing the investment and operation costs while ensuring the acceptable voltage and loading profiles.

Several studies have focused on the design of SCs to enhance the strength of weak grids, the objective function to be minimized typically being the cost of the SCs' installations [11,16–18]. However, when searching for a technical solution under planning studies the costs to acquire the new installations are uncertain as they normally depend upon intricate tendering processes.

Manual iterative procedures to define SCs' ratings and locations were proposed based on WSCR calculations [19] and time-domain simulations [20] but without guaranteeing that an optimal solution is found. In other cases [11,16–18,21,22], a genetic algorithm (GA) was used to solve the optimization problem. However, to obtain a good GA performance, the tuning of its parameters is essential and may consume valuable time during the network planning phase. Besides the GA, in [16] the bat algorithm and the hyper spherical search were used to solve the optimization problem, and the three methods led to different results. In most cases [11,16–18,20–22], the system strength represents a constraint in the optimization process and is quantified by the SCR at the PoCs of the IBRs, thus ignore the potential interactions between IBRs.

This paper proposes a method for optimizing the size and location of SCs to ensure compliance with a minimum ESCR value at a specific set of monitored buses. This method is meant to be used in a decision-making process that involves several other technical and non-technical factors relevant to a more general problem, that of the energy and economic development planning [23].

The case study results reveal that the optimization problem can have multiple technical solutions with values of the objective function within a narrow range around the global minimum. This means that the minimum grid strength constraint can be ensured by sets of SCs having different allocation of individual capacities to the candidate buses but similar total values. Therefore, instead of focusing on finding the single global optimum solution (whose total SC capacity may marginally differ from other solutions), we propose to identify the area of the search space that yields the set of possible solutions. This information is crucial in the planning of large-scale systems, where numerical solutions with differences within a few nominal MVA virtually have the same practical significance.

The proposed methodology to explore the optimization space entails utilizing a basin hopping algorithm (BHA). Due to the exponential increase in the size of the optimization problem relative to the number of candidate and monitored buses, it is proposed to sequentially execute the optimization in runs of increasing level of refinement. In the case studies, the BHA results are validated with the results from a grid search algorithm (GSA) when SCs in two and three candidate buses are optimized.

The main contributions of this paper are as follows:

1. The ESCRs are obtained from short-circuit calculations that are iteratively run so as to cater for the non-linear voltage–current behaviour of the IBRs [24,25].
2. It is proposed to optimize the SC plant subtransient MVA short-circuit contribution at its PoC instead of its nominal power and subtransient reactance, which would be eventually defined by the equipment manufacturer through the joint design of the SC and its step-up transformer.
3. The identification of a range of solutions instead of a single optimum is pivotal for the decision markers since it provides the flexibility to weigh other factors when defining the final specification.
4. A reliability criterion is proposed to rank the solutions based on how evenly the SCs' short-circuit power contributions are distributed among the candidate buses.
5. It is shown that the reduction of the unit costs with the increase of the SC size concentrates the short-circuit contribution in particular buses. Hence, achieving the least costly and the most evenly distributed solutions are opposing goals.

The remainder of this paper is organized as follows. Section 2 outlines the formulation of the optimization problem, while Section 3 presents the proposed solution. Sections 4, 5, and 6 provide the results of the optimization of SCs' sizes in two, three, and all buses, respectively, of a modified IEEE 39-bus benchmark system where several synchronous generators were replaced by IBRs. Finally, Section 7 summarizes the conclusions.

2. Problem formulation

The problem formulation assumes that the planning network model under analysis incorporates future generation and transmission system expansion projects, and its operating point was determined in a separate process from the long-term energy and market optimization.

The ESCR calculation requires to quantify the electrical coupling between the GFL IBRs by means of assessing the voltage change observed at one GFL IBR bus when a voltage change is applied at another GFL IBR bus. To that purpose the interaction factor (IF), also known as multi-infeed interaction factor [26], is defined as

$$IF_{jk} = \frac{\Delta u''_k}{\Delta u''_j}, \tag{1}$$

where $\Delta u''_k$ is the voltage change observed at bus k for a voltage change $\Delta u''_j$ applied at bus j. IBR buses electrically far apart have IF values near zero, whereas for neighbouring buses the IF is close to one ($IF_{jj} = 1$).

The ESCR at the jth bus considering the interaction of N_{IBR} number of GFL IBRs is [5,6,26]

$$ESCR_j = \frac{S''_j}{\sum_{k=1}^{N_{\text{IBR}}} IF_{jk} \cdot P_k}, \tag{2}$$

where P_k is the nominal MVA rating of the GFL IBR connected to the kth bus, and S''_j is the subtransient three-phase short-circuit power at the jth bus. Only the GFL IBR power is to be included in the denominator of (2). This is done because the GFL IBR current injections need external devices that define the voltage waveform, without which the proper GFL IBR operation cannot be guaranteed.

Given a set of N candidate nodes to install SCs with MVA short-circuit contributions S''_i, the problem to be solved is to find the minimum total short-circuit contribution of the SCs that ensures a minimum grid strength $ESCR_{\min}$ at M monitored nodes:

$$\begin{aligned} \min_{\mathbf{S''}} \quad & f_0(\mathbf{S''}) = \sum_{i=1}^{N} S''_i \\ \text{s.t.} \quad & S''_i = 0 \text{ or } S''_{\min} \le S''_i \le S''_{\max} \quad \text{for all} \quad i = 1, \dots, N \\ & ESCR_j \ge ESCR_{\min} \quad \text{for all} \quad j = 1, \dots, M \end{aligned} \tag{3}$$

with $\mathbf{S''} = [S''_1, \dots, S''_N]$. According to the first constraint in (3), an SC is not installed in the ith node if $S''_i < S''_{\min}$. The $ESCR_{\min}$ is a system-specific input to the problem which must be defined in a separate process by means of detailed static and dynamic calculations.

3. Proposed problem solution

This section presents the proposed approach to solve the optimization problem in (3), including the calculation of the ESCR, the optimization algorithms, and the definition of additional criteria to rank the solutions.

3.1. Short-circuit calculations to obtain the ESCR

The short-circuit currents used to calculate both S_j'' and IF_{jk} in (2) are obtained considering the contribution of all rotating resources (synchronous and asynchronous machines) and IBRs [24,27], which are modelled as described below.

3.1.1. Modelling of IBRs

The IBRs with post-fault reactive current injection capability are represented as controlled current sources with the following non-linear current–voltage relationship that accounts for the maximum current ($i_{\max,k}$) limitation [8,24]:

$$\Delta i''_{\mathrm{IBR}_k} = \min\left\{K_k \cdot \Delta u''_k; i_{\max,k}\right\}, \tag{4}$$

where $\Delta i''_{\mathrm{IBR}_k}$ is the current injected by the kth IBR during a short-circuit, and $\Delta u''_k$ is the positive-sequence subtransient voltage change (difference between pre- and post-fault voltages) at the IBR's PoC. The parameter K_k is the gain of the dynamic voltage support (aka "fast fault current") control loop of the inverter, which is set according to the applicable grid code requirements [24,28,29] (typically set between 0 pu and 2 pu); its value is therefore case-specific and can be gathered from the actual equipment data. The voltage $\Delta u''_k$ is also used to calculate the IF in (1).

3.1.2. Modelling of SCs

The ith SC-step up transformer set is modelled as a series reactance x''_i behind a voltage source of magnitude u_i^{Ldf}. The reactance x''_i is defined as

$$x''_i = \frac{\left(u_i^{Ldf}\right)^2}{S''_i}, \tag{5}$$

where u_i^{Ldf} is the steady-state voltage at the PoC of the SC plant obtained from a load flow calculation and S''_i is the parameter optimized by the algorithms described in Section 3. The short-circuit current contribution of the ith SC is therefore

$$i''_i = \frac{u_i^{Ldf} - u''_i}{x''_i}, \tag{6}$$

where u''_i is the subtransient voltage at the SC's PoC during the short-circuit.

3.1.3. Short-circuit calculation procedure

The following three-step approach is applied to calculate the short-circuit currents and voltages [24]:

1. Calculate the short-circuit currents and voltage changes without the contribution of the IBRs.
2. Calculate the current contribution of the IBRs based on the voltage changes calculated in step 1 using (4).
3. Calculate the short-circuit currents with the contribution of the IBRs.

Steps 2 and 3 are iteratively executed until the changes in the voltage deviations and IBR current contributions in two consecutive iterations lie within a (relatively small) predefined range.

3.2. Optimization problem reformulation

As the $ESCR_{\min}$ constraint in (3) is not directly applied to the optimized variables in $\mathbf{S}''$, it is proposed to embed it into the objective function. Hence, the objective function f_0 is changed to $f_1 : \mathbb{R}^N \to \mathbb{R}$ as follows:

$$\begin{aligned} &\min_{\mathbf{S}''} \quad f_1(\mathbf{S}'') = \sum_{i=1}^{N} S''_i + ESCR_{pen} \\ &\text{s.t.} \quad S''_i = 0 \text{ or } S''_{\min} \le S''_i \le S''_{\max} \quad \text{for all} \quad i = 1, \dots, N \end{aligned} \tag{7}$$

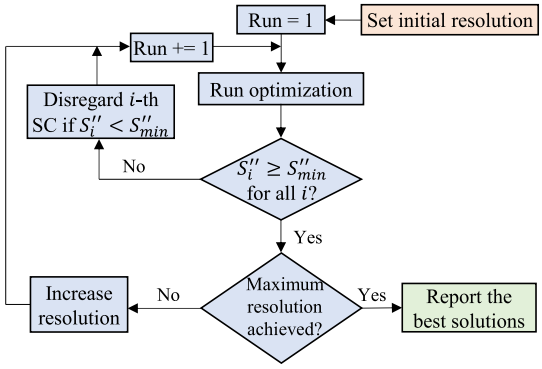


Fig. 1. Iterative execution of the optimization algorithms. The resolution is increased by reducing $(b_i - a_i)$ and r_i in the GSA, and d in the BHA.

with

$$ESCR_{pen} = \begin{cases} \gamma & \text{if } ESCR_j < ESCR_{\min} \quad \text{for all} \quad j = 1, \dots, M \\ 0 & \text{otherwise} \end{cases} \tag{8}$$

$$\gamma = 10^6 \cdot \max\left\{ESCR_{\min} - ESCR_j\right\} \tag{9}$$

The $ESCR_{pen}$ component of f_1 strongly penalizes the combinations $\mathbf{S}''$ in which the minimum ESCR among all monitored buses is below $ESCR_{\min}$. In practice, it prevents invalid solutions while still providing information to the optimization algorithm about which changes in f_1 tend to increase the values of ESCR.

3.3. Optimization of the size and location of the SCs

This section provides an overview of the approach proposed to solve the optimization problem formulated in Section 3.2 by means of the BHA. The GSA is also described but only used in the case studies to validate the BHA results when applied for the optimization of two and three SCs.

An SC connected at an IBR's PoC mainly helps to increase the ESCR locally when other IBRs' PoCs are electrically far. However, if an SC is properly located somewhere between two IBRs, the ESCRs of both IBRs' PoCs could be improved to some extent [22]. Therefore, the set of candidate nodes should include nodes in addition to the IBR's PoCs.

The following checks are performed before starting the optimization:

1. To ensure that the problem has a solution, it is initially verified that $ESCR_j \ge ESCR_{\min}$ when $S_i = S_{\max}$ for all i and j. Otherwise, other candidate nodes must be added or, alternatively, $S_{\max}$ must be raised.
2. To reduce the analysis only to the weak nodes of the grid, it is initially verified that $ESCR_j^0 < ESCR_{\min}$, where $ESCR_j^0$ is the ESCR of the jth monitored node in the scenario without SCs. All nodes in which this condition is not met are removed from the list of monitored nodes.

The search space can be initially large and the initial guess $\mathbf{S}''_{ini}$ of the solution far from the optimum. Therefore, it is proposed to approach the area of the global optimum by executing the optimization algorithm (either GSA or BHA) in runs of increasing resolution, as shown in Fig. 1. The resolution refers to the granularity of the search, i.e. indicates how accurately the algorithm approaches the solution. A higher algorithm resolution increases the precision of the result and the required computational effort.

In the remainder of this paper an algorithm run refers to one loop of the workflow in Fig. 1. An algorithm iteration refers to one internal calculation of the algorithm itself for the same run. After each run the nodes for which $S''_i < S''_{\min}$ are disregarded (no longer considered as candidate nodes) so as to fulfil the constraint in (7).

3.3.1. Grid search algorithm

The GSA comprises the exhaustive search over the parameter space for discrete values of the parameters within a predefined range. For each combination of parameters (grid points) the algorithm evaluates the objective function, the optimum solution being the combination that leads to the lowest value.

The search space is defined by the vectors $\mathbf{a} = [a_1, a_2, \dots, a_N]$ and $\mathbf{b} = [b_1, b_2, \dots, b_N]$ of lower and upper bounds, respectively, for each element of $\mathbf{S}''$. The search vectors comprise P equally spaced points in each range $\mathbf{R}_i = [a_i, b_i]$, including a_i and b_i. Considering that all parameter spaces are identical, the grid step is $r_i = (b_i - a_i)/P$. The resolution of the GSA (see Fig. 1) is increased by reducing $(b_i - a_i)$ and r_i.

The main drawback of the GSA is that the total number of grid points to be evaluated is P^N, which is prohibitive in practical (high dimension) applications due to the high computational burden. In those cases, a more efficient algorithm must be applied, like the BHA described next.

3.3.2. Basin hopping Monte Carlo optimization

Instead of the GSA, it is proposed to use the BHA to screen the continuous space of objective function values by hopping among different regions around local minima (basins). This is achieved by random Monte Carlo moves followed by local minimization to which an acceptance/rejection criteria is applied. The local exploration (minimization) is performed by a Nelder–Mead simplex algorithm (NMA) [30], which can converge to a local minimum or a non–stationary point [31,32], hence its result depends upon its initialization (initial simplex). In Fig. 1, a BHA run comprises several iterations, each of which corresponds to an NMA run.

Eventually, the local minima from each BHA iteration are collected to identify the area in which optimal solutions within a narrow range are located. In the case studies of this paper the BHA and NMA implementations from the *scipy* Python package [33] were used. An overview of both methods is presented below.

Basin hopping algorithm

The BHA is a global optimization algorithm specially suited for non-linear objective functions in high-dimensional landscapes with multiple local optima [34,35]. It combines heuristic procedures with local searches to enhance its exploration of the given variable space [36].

The BHA consists of an iterative process starting from a candidate solution $\mathbf{S}''_{ini}$ where another search algorithm is applied to find a local minimum. Then, the BHA performs random perturbations of maximum size d on the value of each variable found in the previous iteration to jump to a new region in the search space, where the local search algorithm is applied again [36]. The current minimum is accepted and retained in memory if it is lower the lowest value obtained for the objective function so far. Instead of letting the BHA converge to a single optimum solution, we propose to run it for a number of iterations [36] sufficiently large that allows to identify the area of the search space in which the set of optimum practical solutions is located. The resolution of the BHA (see Fig. 1) is increased by reducing d.

Nelder–Mead simplex algorithm

The NMA is a direct search method that attempts to minimize a scalar-valued non-linear objective function of N real variables using only function values. It maintains at each step a non-degenerate simplex, which is a geometric figure in N dimensions of non-zero volume that is the convex hull of $N + 1$ vertices. Each iteration begins with a simplex, specified by its $N + 1$ vertices and the associated function values. One or more test points are computed, along with their function values, and the iteration terminates with a new (different) simplex such that the function values at its vertices satisfy a descent condition compared to the previous simplex [37]. The NMA optimization terminates when the vector distance moved in that step is fractionally smaller in magnitude than some tolerance *xatol*. Alternatively, it can be required the decrease in the function value in the terminating step to be fractionally smaller than some tolerance *fatol* [38].

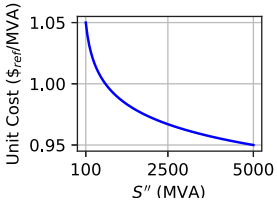


Fig. 2. SC unit cost $c(S'')$.

The NMA possesses two distinctive features that render it highly appropriate for our present problem: low computational cost and robustness. Its derivative-free nature makes it fast and straightforward to implement. Additionally, its robustness guarantees reliable performance across a broad range of application scenarios. Despite only achieving linear convergence, the algorithm's performance is satisfactory for our objectives. Rather than a highly specialized solver that marginally outperforms the NMA, we prioritize a dependable solver that can be utilized in diverse problem domains.

3.4. Additional criteria to rank the solutions

The case study results in Sections 4, 5 and 6 indicate that the problem in (7) can have several solutions with similar values of the objective function (these values are identical in the GSA). Therefore, this section proposes two criteria to rank several solutions based on the total equipment cost and a reliability index, which are calculated by post-processing the optimization results. Excluding these criteria from f_1 allows to change them at a later stage without the need to rerun the optimization.

3.4.1. Selection of the most evenly distributed combination

It is proposed to rank the solutions according to the extent to which the SCs' short-circuit contributions are evenly allocated among the candidate buses. This attribute is quantified by the following index:

$$e = \frac{1}{\sigma_{\mathbf{S}''}}, \tag{10}$$

where $\sigma_{\mathbf{S}''}$ denotes the standard deviation of the elements of $\mathbf{S}''$. For example, with a low value of $\sigma_{\mathbf{S}''}$, the N vector elements are closer to the set's mean, resulting in a more even distribution of short-circuit contributions.

Combinations with lower e concentrate the short-circuit contribution in a reduced number of candidate nodes, thus tend to be less reliable because the failure of the largest SC increases the negative impact on the system strength. The higher e (lower $\sigma_{\mathbf{S}''}$), the more evenly distributed the short-circuit contributions are among the candidate nodes.

3.4.2. Selection of the least costly solution

Due to the economy of scale [39], the cost of an SC typically tends to be reduced with the increase of its size. This cost differentiation can be used to select the least expensive solution. For illustration purposes, the case studies of this paper assume the non-linear cost function in (11) with the unit cost function $c(S'')$ in Fig. 2 expressed in terms of a reference cost $\$_{ref}$ per MVA.

$$C(\mathbf{S}'') = \sum_{i=1}^{N} c(S''_i) \cdot S''_i \quad \text{with} \quad c(S'') = 1.181 \cdot \left(S''\right)^{-0.026} \tag{11}$$

4. Case study A: Optimization of two SCs

The GSA and BHA were applied to define the optimum SCs that guarantee an $ESCR_{\min} = 1.5$ at all buses of a modified 39-bus New England system model [40]. In order to produce weak network conditions, the six synchronous generators of the original model connected to

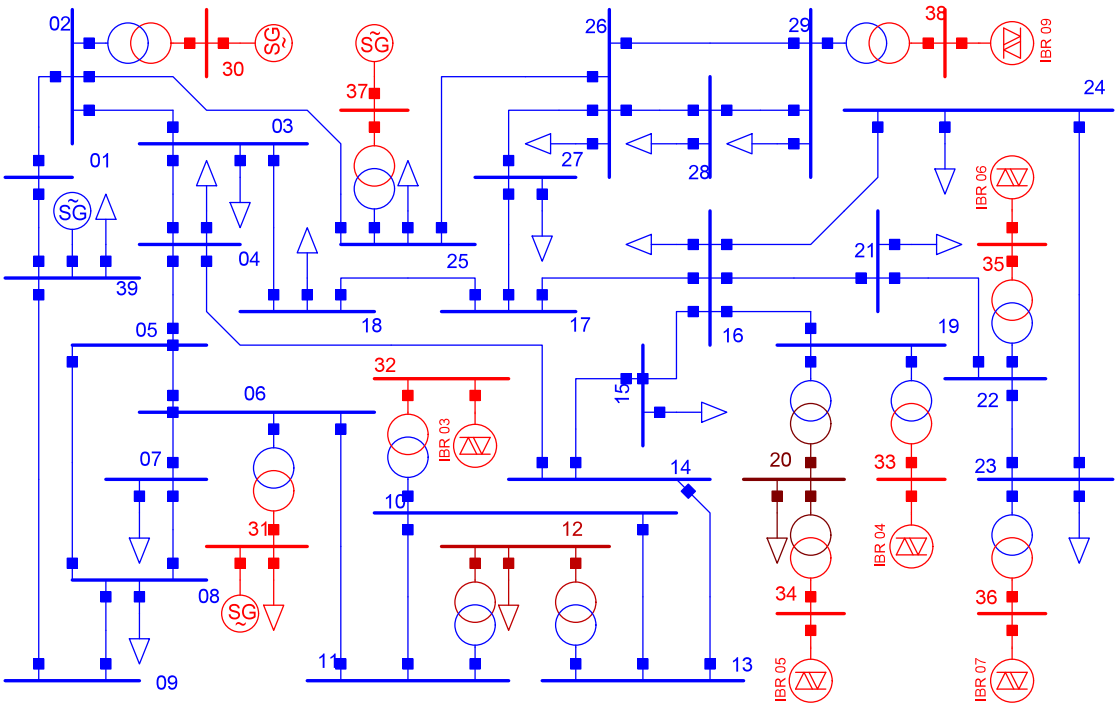


Fig. 3. Modified IEEE 39-bus benchmark system with IBRs in buses 32 to 36 and 38.

Table 1
GSA search space definition in case study A1 (optimization of two SCs).

GSA	Range **R** (MVA)		r
run	SC21	SC28	(MVA)
1	[0,10 000]	[0,10 000]	100
2	[4800,5700]	[300,1200]	5

buses 32 to 36 and 38 were arbitrarily replaced by GFL IBRs of identical nominal power and MW dispatch, as shown in Fig. 3. In the analysed scenario, 62% of the total 6144 MW generation is produced by IBRs.

Buses 21 and 28 were defined as candidates for the installation of the SCs SC21 and SC28, respectively. The IBR parameters K_k and $i_{\max,k}$ in (4) were both set to 1 pu for all k. DIgSILENT PowerFactory 2022 software was used to represent the system model and run the electrical calculations.

4.1. Case study A1: Grid search results

The GSA was applied as per Fig. 1 in two runs having the parameters in Table 1. Run 2 was centred around the global minima obtained in run 1.

Fig. 4 shows the value of f_1 for all combinations $\mathbf{S}''$ in the GSA run 1. The minimum ESCR is met in 6045 out of 10 201 grid points. There are 8 combinations leading to the minimum value of the objective function of 6000 MVA (crosses in Fig. 4). In these combinations, the short-circuit contributions of SC21 and SC28 lie within 4900 and 5600 MVA, and 400 and 1100 MVA, respectively. For instance, the solutions $[S''_{SC21}, S''_{SC28}]$ are given by the pairs [4900,1100] MVA, [5000,1000] MVA, and so on.

Fig. 5 shows the SC combinations leading to $f_1 \leq 6000$ MVA that fulfil the minimum ESCR requirement in run 2. The global minimum value of 5935 MVA is met by 30 solutions in which the short-circuit contributions range from 5160 to 5305 MVA for SC21 and from 630 to 775 MVA for SC28. Therefore, in run 2 the value of the global minima is reduced and the ranges of short-circuit contributions are narrowed when compared to run 1.

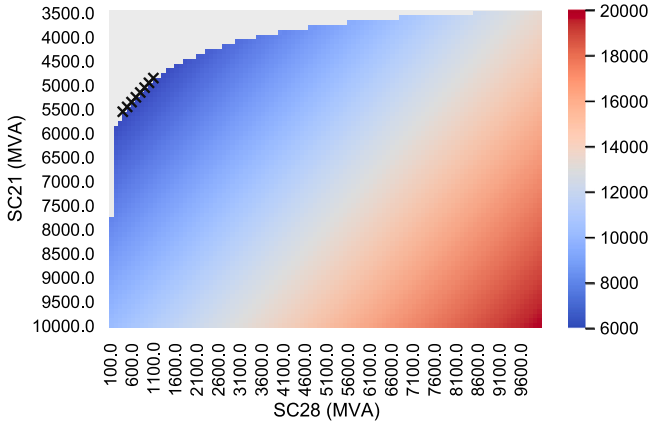


Fig. 4. Values of f_1 (in MVA) for the GSA run 1 (r = 100 MVA) in case study A1. The values in the grey area (top left) do not meet the minimum ESCR constraint. The crosses indicate the 8 global minima.

4.2. Case study A2: Basin hopping results

The BHA was ran only once considering $S''_{\min}$, $S''_{\max}$ and S''_{ini} equal to 0 MVA, 10 000 MVA and 5000 MVA, respectively, for all SCs. The maximum numbers of NMA and BHA iterations were set equal to 500 and 10, respectively, d = 500 MVA and $xatol$ = $fatol$ = 0.1 MVA. In order to be able to reproduce the results in different calls, a fixed seed was used to define the random change of the starting point of each BHA run.

Table 2 and Fig. 6 show the solutions of each BHA iteration. Iteration 1 already provides a solution with a value of f_1 close to the minimum found by the GSA. This solution leads to the highest values of f_1 and C but the most evenly distributed solution (highest e). Iterations 5 and 2 lead to the least evenly distributed and least costly solutions, respectively.

It is verified in Fig. 7 that the solutions of the BHA iterations lie in the region around the global optimum obtained from the GSA

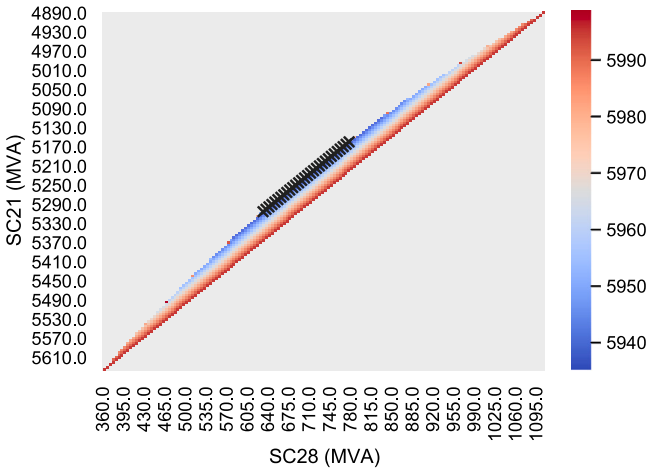


Fig. 5. Global optima (×) for the GSA run 2 ($r = 5$ MVA) in case study A1. The values in the top grey area do not meet the minimum ESCR constraint, whereas the values in the bottom grey area are higher than 6000 MVA.

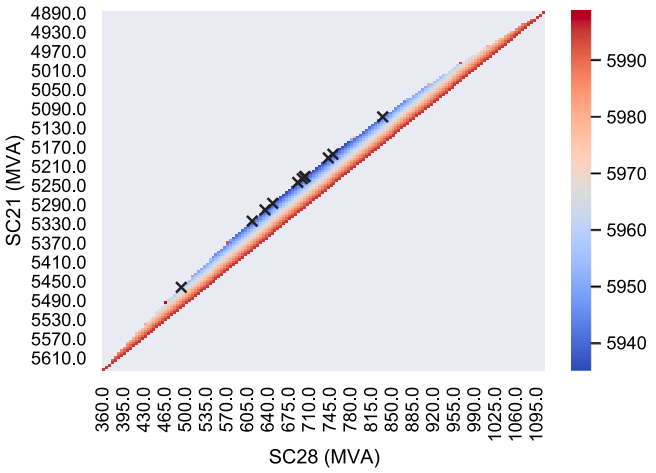


Fig. 7. Solutions of the BHA iterations in case study A2 marked with crosses (×) on top of the grid search evaluations (heat map) around the global optima in case study A1.

Table 2
BHA solutions of case study A2 (optimization of two SCs).

BHA iteration	SC21 (MVA)	SC28 (MVA)	f_1 (MVA)	e (1/GVA)	C ($\$_{ref}$)
1	5108	832	5940	0.468	5677.23
2	5287	647	5934	0.431	5663.04
3	5185	748	5933	0.451	5667.63
4	5193	740	5933	0.449	5666.99
5	5324	612	5936	0.424	5663.42
6	5301	634	5935	0.429	5663.06
7	5232	700	5933	0.441	5664.44
8	5231	701	5933	0.441	5664.49
9	5236	696	5933	0.441	5664.27
10	5244	689	5933	0.439	5663.95

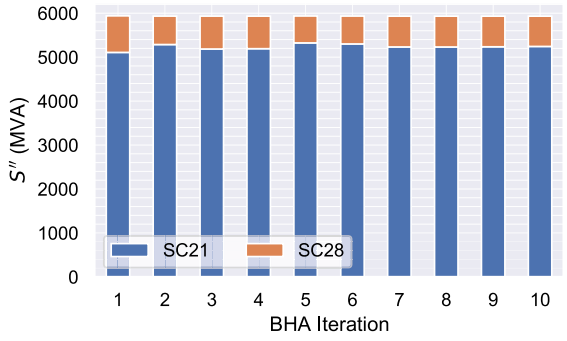


Fig. 6. BHA solutions in case study A2 (optimization of two SCs).

calculations in Section 4.1 (see Fig. 5). The minimum value of the objective function (5933 MVA) is 0.042% lower than the one obtained in the GSA run 2.

5. Case study B: Optimization of three SCs

The GSA and BHA were applied to the modified IEEE 39-bus system considering bus 14 as a third candidate node for the installation of an SC.

5.1. Case study B1: Grid search results

The GSA was executed in three runs of increasing refinement as per Fig. 1, with the search space parameters in Table 3. In this case, the same minimum value of the objective function is obtained in runs 2 and 3, which indicates that the global minimum is found already in run 2.

Table 3
GSA search space definition in case study B1 (optimization of three SCs).

GSA run	Range **R** (MVA)			r (MVA)	Optima	
	SC21	SC28	SC15		Number	(MVA)
1	[0,10 000]	[0,10 000]	[0,10 000]	200	38	5600
2	[2600,4350]	[200,950]	[800,2750]	50	16	5450
3	[3300,3545]	[300,395]	[1550,1795]	5	898	5450

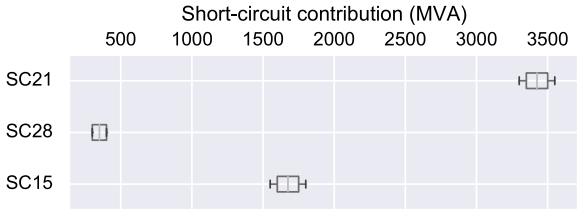


Fig. 8. Ranges of short-circuit contributions obtained in GSA run 2 of case study B1.

Fig. 8 shows the ranges of the short-circuit contributions of the three SCs for all solutions of the GSA run 2. SC21 and SC15 are connected electrically closer to five IBRs' PoCs, thus have wider ranges of short-circuit contributions than SC28. SC28 has a more local effect and mainly increases the strength at the IBR09 mains (bus 38), hence its tighter range.

As an example, Table 4 shows six selected solutions of the GSA run 2 ($r = 50$ MVA) with different values of e and C, all of which have the same value of f_1. Solution 6 has the lowest values of both e and C whereas the opposite applies to solution 1.

5.2. Case study B2: Basin hopping results

The BHA was configured as in case study A2. Table 5 and Fig. 9 show the solutions found in each BHA iteration. The BHA converges relatively fast since the solution of iteration 4 is already very close to the optimum found by the GSA. The minimum value (5447.1 MVA) is obtained in iteration 10 for which Fig. 10 shows the NMA convergence. Iterations 3 to 10 represent different combinations of SC sizes, all of them leading to values of f_1 within 1% difference from the overall

Table 4
Solutions of GSA run 2 with different values of e in case study B1.

GSA solution	SC21 (MVA)	SC28 (MVA)	SC15 (MVA)	f_1 (MVA)	e (1/MVA)	C ($\$_{ref}$)
1	3300	400	1750	5450	0.844	5281.63
2	3300	350	1800	5450	0.830	5279.70
3	3400	400	1650	5450	0.813	5279.99
4	3450	350	1650	5450	0.787	5277.27
5	3500	350	1600	5450	0.772	5276.35
6	3550	300	1600	5450	0.749	5273.41

Table 5
BHA solutions in case study B2 (optimization of three SCs).

BHA iteration	SC21 (MVA)	SC28 (MVA)	SC15 (MVA)	f_1 (MVA)	e (1/GVA)	C ($\$_{ref}$)
1	5216	710	4	5930	0.433	5663.51
2	5267	666	0	5933	0.427	5663.27
3	3453	656	1379	5488	0.844	5322.02
4	3468	458	1527	5453	0.803	5283.33
5	3341	181	1944	5465	0.773	5285.26
6	3495	179	1788	5461	0.739	5279.35
7	3416	562	1488	5467	0.841	5300.34
8	3446	301	1701	5448	0.777	5273.16
9	3415	433	1602	5450	0.815	5280.60
10	3422	316	1709	5447	0.787	5273.68

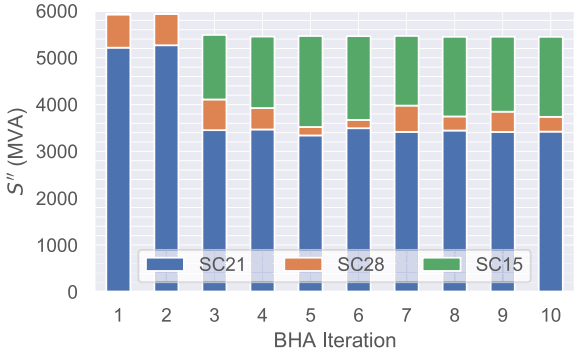


Fig. 9. BHA solutions in case study B2 (optimization of three SCs).

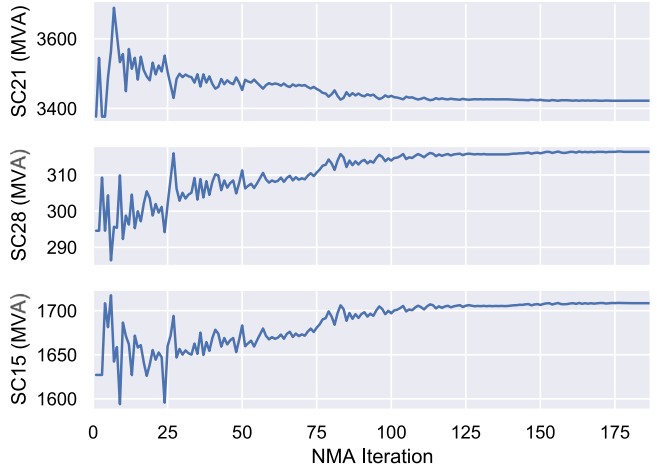


Fig. 10. NMA convergence of BHA iteration 10 in case study B2.

minimum. The most and least evenly distributed combinations are obtained in iterations 3 and 2, respectively, whereas the most and least costly combinations are obtained in iterations 1 and 8, respectively.

Table 6
Solutions of the four BHA runs in case study C (SCs in all buses).

BHA run	f_1 (MVA)	Improvement[a] (%)	Candidate nodes N
1	3414.6	–	39
2	2627.3	23.1%	25
3	2433.7	7.4%	13
4	2414.4	0.8%	6

[a] With respect to the previous run.

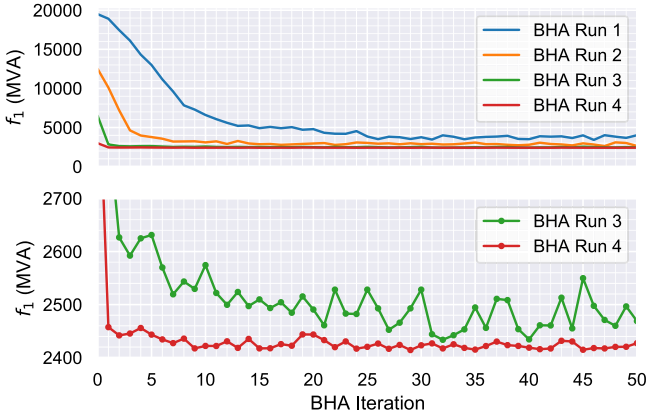


Fig. 11. Convergence of the four BHA runs in case study C (top) and enlarged view of runs 3 and 4 (bottom). Iteration 0 represents the starting point.

6. Case study C: Global search

The BHA was applied to optimize the size and location of SCs considering the 39 network buses as candidates for the installation of SCs. The use of the GSA was disregarded in this case because its calculation time would be excessively high.

Since the search space is initially relatively large, the BHA was iteratively run following the workflow in Fig. 1 with $S''_{\min} = 50$ MVA. The first run included all network buses as candidate nodes. Then the best three solutions were identified and the buses with S'' allocation below $S''_{\min}$ in the three solutions were discarded for the second run. Using more than one solution avoids discarding candidate nodes which can potentially improve the objective function if a different combination of SCs was eventually selected. The same procedure was applied in the subsequent runs until the best solution could only be marginally improved ($\leq$1%).

The configuration of the BHA was modified compared to the previous case studies. The number of BHA iterations per run was increased from 10 to 50 to obtain a sufficiently large number of solutions in the area of the global minimum, and d was reduced from 500 MVA to 100 MVA. In all runs S''_{ini} was set to 500 MVA for all candidate nodes.

Table 6 shows the best solution found in each BHA run, including the number of candidate nodes considered. Fig. 11 shows the value of the objective function for each iteration of the BHA runs. In the BHA run 4 the values of f_1 in all NMA solutions are already less than 2% higher than the overall minimum (2414.4 MVA). All additional iterations beyond these points allow to further screen the search space to identify other solutions equally valid in practice.

Fig. 12 shows the distribution of the short-circuit contributions of the six candidate buses. On the one hand, the buses with the lowest potential for the installation of SCs are buses 20 and 36. On the other hand, the required contribution at bus 29 (SC29) lies within a relatively narrow range around 365 MVA.

Table 7 presents the most and least evenly distributed and costly combinations of short-circuit contributions in the BHA run 4. The BHA

Table 7
Most and least evenly distributed and costly solutions of the BHA run 4 in case study C.

NMA run	SC19 (MVA)	SC20 (MVA)	SC22 (MVA)	SC23 (MVA)	SC29 (MVA)	SC36 (MVA)	f_1 (MVA)	e (1/GVA)	C ($\$_{ref}$)
19	885	161	529	422	367	79	2443.80	3.81	2462.18
39	1019	0	408	627	367	0	2422.12	2.82	2423.58
4	882	152	307	743	365	6	2455.86	3.23	2466.60
29	1027	0	490	529	368	0	2414.45	2.85	2416.42

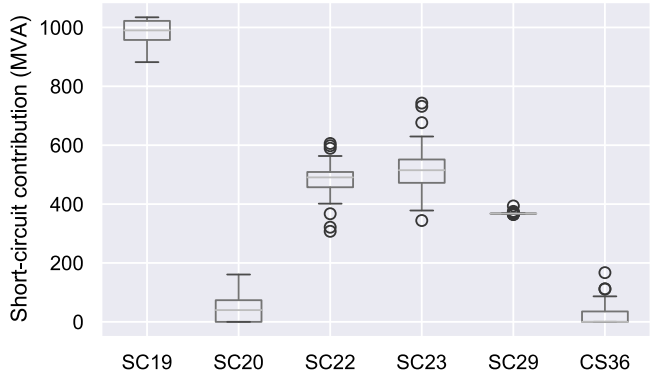


Fig. 12. Distribution of the short-circuit contributions for the six candidate buses in the BHA run 4 of case study C.

iterations 19 and 39 lead to the most and least evenly distributed combinations, respectively, whereas the most and least costly combinations are obtained in iterations 4 and 29, respectively.

7. Conclusions

The proposed approach allows to automatically identify multiple solutions that guarantee compliance with a minimum grid strength requirement. These solutions lead to approximately the same total SC capacity but different allocation to the candidate buses. The identification of such solutions provides decision makers with an additional degree of freedom to consider other criteria (e.g. total inertia or minimum commissioning time) at the time of selecting the final solution.

The solution leading to the most evenly distributed short-circuit power among all SCs can be identified as the one having the lowest standard deviation of the individual contributions. However, the economy of scale, resulting in a reduction in the unit cost of the SCs with an increase in their size, can lead to solutions less evenly distributed and less reliable.

The primary limitation of the proposed method lies in its dependence on the ESCR to identify potential grid strength issues. Given that the ESCR is heuristic, the minimum value of ESCR considered for the optimization should be calculated by means of detailed static and dynamic analyses and embed a safety margin.

Future research should include the comparison of the performance and accuracy of the BHA to that of other optimization algorithms and the assessment of other grid strength indexes. While this paper focused solely on optimizing the SCs' design, the method should eventually be integrated to a broader approach that also optimizes other grid strengthening solutions, such as the installation of GFM IBRs, coordinated with the optimization of the generation and transmission expansion.

CRediT authorship contribution statement

Fernando De Marco: Methodology, Software, Formal analysis, Writing – original draft. **José Gómez:** Conceptualization, Validation, Visualization, Writing – review & editing. **Valentin Bolz:** Conceptualization, Investigation, Writing – review & editing. **Flavio Fernández:** Supervision, Project administration.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Fernando De Marco reports financial support was provided by DIgSILENT GmbH. Jose Gomez reports financial support was provided by DIgSILENT GmbH. Valentin Bolz reports financial support was provided by DIgSILENT GmbH. Flavio Fernandez reports financial support was provided by DigSILENT GmbH. Fernando De Marco reports a relationship with DIgSILENT GmbH that includes: employment. Jose Gomez reports a relationship with DIgSILENT GmbH that includes: employment. Valentin Bolz reports a relationship with DIgSILENT GmbH that includes: employment. Flavio Fernandez reports a relationship with DIgSILENT GmbH that includes: employment.

Data availability

No data was used for the research described in the article.

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